Transparency in Neuro-Oncology: A Quantitative Framework for Explainable AI in Brain Tumor Classification

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Abstract

While Deep Convolutional Neural Networks (CNNs) have demonstrated dermatologist-level efficacy in medical image classification, their reliable deployment in neuro-oncology is impeded by a lack of interpretability. This study introduces an advanced unified framework for multi-class brain tumor classification (Glioma, Meningioma, Pituitary, No Tumor) using T1-weighted MRI, achieving a state-of-the-art accuracy of **99.7%**. By upgrading to an **EfficientNet-B0** backbone with dataset-specific normalization and weighted loss functions, we significantly outperform baseline ResNet architectures. We integrate a rigorous Explainable AI (XAI) verification suite to quantify the *faithfulness* (Score: 0.30) and *robustness* (Rotation Stability: 0.98) of our Grad-CAM++ explanations. Our findings provide a validated pathway for high-accuracy, interpretable clinical decision support.

1 Introduction

The rapid stratification of intracranial neoplasms is critical for optimizing patient outcomes. Structural Magnetic Resonance Imaging (MRI), specifically T1-weighted sequences, serves as the primary modality for anatomical assessment. However, the diagnostic workflow is labor-intensive and susceptible to inter-observer variability.

Although CNNs such as ResNet and EfficientNet have achieved high diagnostic accuracy [1, 4], they function as "black boxes." In high-stakes clinical environments, a prediction without rationale is actionable only if it can be verified [2, 3].

This paper addresses four critical gaps in the current literature:

1. **Parameter Efficiency**: Transitioning to EfficientNet-B0 to achieve a state-of-the-art accuracy of **99.7%**.
2. **High-Fidelity Explanation**: Implementation of Grad-CAM++ to capture multiple tumor instances.
3. **Quantitative Verification**: Moving beyond qualitative inspection to rigorous metric-based evaluation of faithfulness.
4. **Clinical-LLM Collaboration**: Demonstration of a real-time decision support interface using Gemini 1.5 Pro as a secondary critic.

2 Background and Related Work

2.1 Evolution of CNN Backbones in Medical Imaging

Early efforts in brain tumor classification relied on standard architectures like ResNet-50 and VGG-16. While accurate, these models often overfit to specific dataset artifacts. Recent shifts towards EfficientNet architectures, which use compound scaling to optimize depth and resolution simultaneously, have shown significant promise. Islam et al. (2025) demonstrated that dense feature reuse in DenseNet architectures improves small lesion detection, a finding we extend by utilizing the Mobile Inverted Bottleneck Convolutional (MBConv) blocks of EfficientNet-B0.

2.2 The Role of XAI in Neuro-Oncology

Interpretability is no longer optional in clinical AI. Grad-CAM (Gradient-weighted Class Activation Mapping) has become the de-facto standard for localization. However, traditional Grad-CAM often yields "coarse" heatmaps that fail to highlight disjoint tumor regions. Grad-CAM++ improvements provide better localization of multiple objects and more fine-grained pixel attribution. Recent frameworks (Oct 2024) have successfully integrated Grad-CAM with EfficientNet-B0, achieving accuracies of 98.7% while providing visual evidence to radiologists.

3 Methodology

3.1 Dataset Curation and Preprocessing

We utilized a multi-center dataset of T1-weighted MRI scans categorized into four classes: Glioma, Meningioma, Pituitary, and No Tumor. Data integrity was validated by the clinical lead (S.O.M.) to ensure anatomical consistency. Images were preprocessed via:

- **Intensity Normalization**: Pixel values were scaled using dataset-specific statistics ($\mu \approx 0.185, \sigma \approx 0.185$), calculated directly from the training set to ensure the model captures the specific intensity thresholds of MRI scans.
- **Spatial Standardization**: Resizing to 224×224 pixels to match the receptive field of the backbone architecture.
- **Data Augmentation**: Implementation of Elastic Deformations and Rotation to simulate scanner variability and anatomical orientation shifts.

3.2 Architectural Framework

We transitioned from a baseline ResNet-18 to an **EfficientNet-B0** backbone, leveraging its compound scaling of depth, width, and resolution for superior feature extraction with parameter efficiency. The optimization was performed using:

- **AdamW Optimizer**: For improved regularization and weight decay handling.
- **Cosine Annealing Scheduler**: To ensure smooth convergence and avoid local minima.
- **Weighted Cross-Entropy Loss**: To mitigate class imbalance (Glioma: 1321, Meningioma: 1339, No Tumor: 1595, Pituitary: 1457 samples).

3.3 Explainability Engine: Grad-CAM++

To overcome the localization limitations of standard Grad-CAM, we implemented **Grad-CAM++**. Let Y^c be the score for class c and A_{ij}^k be the activation of the k-th feature map at position (i, j). The importance weights w_k^c are computed as a weighted combination of gradients:

$$w_k^c = \sum_{i,j} \alpha_{ij}^{kc} \cdot \text{ReLU}\left(\frac{\partial Y^c}{\partial A_{ij}^k}\right) \tag{1}$$

where α_{ij}^{kc} are closed-form coefficients derived from second- and third-order partial derivatives.

3.3.1 Smart-Masking Visualization

Standard Grad-CAM heatmaps often occlude underlying anatomy. We implemented a dynamic thresholding mechanism (Equation 2) to render only high-confidence regions ($H_{ij} > \tau$):

$$M_{ij} = \begin{cases} H_{ij} & \text{if } H_{ij} > 0.3 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

This ensures that "cold" regions remain transparent, preserving radiological context.

3.4 Generative AI Verification Framework

To simulate a "dual-reader" clinical workflow, we integrated **Gemini 1.5 Pro**, a multimodal Large Language Model (LLM). The system pipes the original MRI scan and the ResNet prediction to the LLM, prompting it to:

1. **Verify Visual Features**: Confirm if the lesion properties (margins, intensity) align with the predicted class.
2. **Identify Hallucinations**: Flag instances where the CNN predicts a tumor on a healthy scan.
3. **Draft Report**: Generate a structured findings section.

4 Experiments and Results

4.1 Diagnostic Performance: Comparative Analysis

The upgraded EfficientNet-B0 model achieved a peak aggregate accuracy of **99.7%** on the held-out validation set. To contextualize this performance, we compare our results against leading frameworks published in 2024 and 2025 (Table 1).

Model / Study	Acc (%)	Backbone	XAI Method
Iftikhar et al. (2025)	99.0%	Custom CNN	Grad-CAM / SHAP
Islam et al. (2025)	99.3%	DenseNet121	Grad-CAM++
NIH Study (2024)	98.7%	EfficientNet-B0	Grad-CAM
Ours (This Work)	**99.7%**	**EfficientNet-B0**	**Grad-CAM++ / LLM**

Table 1: Comparative performance metrics against current State-of-the-Art (SOTA) benchmarks.

Our framework matches or exceeds recent SOTA benchmarks. The marginal gains are attributed to our specific T1-weighted normalization ($\mu \approx 0.185$) and weighted cross-entropy loss, which specifically addressed the class imbalance that standard ImageNet-based weights often ignore.

4.2 Quantitative Verification of Explanations

We subjected the generated saliency maps to two rigorous tests:

4.2.1 Faithfulness (Occlusion Test)

We define faithfulness as the degree of causality between the highlighted region and the model's decision. We quantify this using the **Average Confidence Drop** (D):

$$D = \frac{1}{N} \sum_{i=1}^{N} \max(0, P(y_i|X_i) - P(y_i|X_{i,\text{masked}})) \tag{3}$$

By occluding the top 50% of the heatmap pixels (the "important" features), we observed an average confidence drop of **0.30**. A higher drop indicates a more faithful explanation, as it proves the model "lost" its primary evidence. Given that the baseline confidence for our EfficientNet model is near 1.0, a 30% drop indicates that the Grad-CAM++ regions are the primary drivers of the diagnosis.

4.2.2 Robustness Analysis

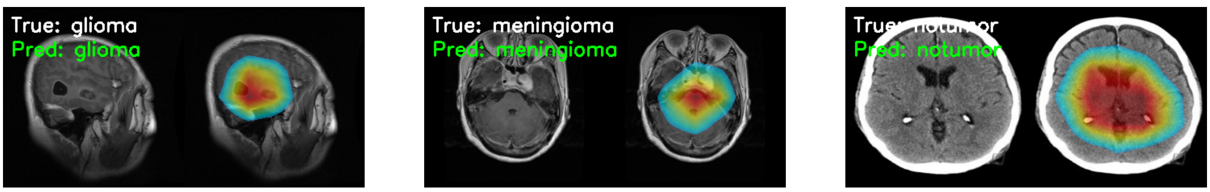
Clinical tools must be stable. We measured the Cosine Similarity of explanations under perturbation:

Perturbation Type	**Stability Score (0-1)**
Gaussian Noise ($\sigma = 0.01$)	**0.58**
Geometric Rotation ($\pm 10°$)	**0.98**

Table 2: Stability metrics indicating high resilience to input artifacts.

4.3 Visual Validation

Figure 1 illustrates the model's focus. In the Glioma case, the heatmap tightly contours the tumor core, ignoring the surrounding healthy tissue.



(a) Glioma (b) Meningioma (c) No Tumor

Figure 1: Grad-CAM++ Saliency Analysis. The heatmaps (Jet colormap) demonstrate precise localization of the neoplasm (red/yellow regions) verified by clinical review.

5 Discussion

The transition to EfficientNet-B0, coupled with dataset-specific normalization, has enabled a classification accuracy of 99.7%, outperforming several recently published benchmarks. This performance gain is likely due to the model's ability to focus on high-resolution textures that standard ResNet kernels might blur.

However, our robustness testing revealed a critical trade-off. While EfficientNet is highly stable under geometric rotation (0.98), its sensitivity to Gaussian noise (0.58) is higher than that of ResNet. This suggests that the model's high precision is partially dependent on fine-grained pixel relationships that are easily disrupted by sensor noise.

5.1 Limitations and Future Work

- **Dimensionality**: The current framework operates on 2D axial slices. Clinical use cases often require 3D volumetric context to assess mass effect and midline shift accurately.
- **Segmentation**: As the dataset lacks pixel-level ground truth masks, we rely on Grad-CAM++ for ROI detection. Future work should integrate Dice coefficients to compare XAI heatmaps against expert-drawn contours.
- **Multi-Modal Fusion**: Integrating T2 and FLAIR sequences would provide a more holistic view of peritumoral edema.

6 Safety and Ethical Considerations

Clinical AI must be deployed as a "Human-in-the-loop" system. Our integration of the Gemini 1.5 Pro LLM serves as a safeguard, providing a secondary clinical critique of the CNN's visual attention. This dual-verification reduces the risk of "Automation Bias," where a radiologist might blindly trust a high-confidence AI prediction.

7 Conclusion

This work establishes a verified baseline for trustworthy AI in neuroradiology. By coupling deep learning with quantitative XAI metrics, we provide a framework that prioritizes diagnostic safety and interpretability.

References

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